Srijan Tiwari

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SUMMARY

Final-Year B.Tech Undergraduate at **IIT Guwahati** with software engineering experience leading a **6-member** development group at **Genuity IO**. Strong foundation in **C++20**, **Python**, low-latency algorithms, cryptographic protocols, and full-stack web architectures across **100,000+** record workloads.

EDUCATION

Indian Institute of Technology (IIT), Guwahati — *Bachelor of Technology (B.Tech)* **Aug 2023 - May 2027** | *Guwahati, India*
*Board Examinations: CBSE Class XII: **84.4%** | CBSE Class X: **94.6%**.*

EXPERIENCE

Genuity IO — *Lead Software Engineer, Synthetic Data Division* **May 2026 - Jul 2026** | *Bengaluru, India*

- **Architected** Genuity **v2.0**, an enterprise synthetic dataset generation library supporting **5 modalities** (tabular, time-series, financial, relational, and text) and **20+ generative models**, driving core production integration into the Vulpix platform.
- **Led** a cross-functional group of **6 developers** across architecture design, code reviews, and Agile sprint delivery; established a benchmarking framework to validate statistical fidelity, utility, and confidentiality across **20+ models** and **11 public benchmarks**.
- **Engineered** production backend workflows for schema profiling, constraint validation, and differential privacy using **Python, FastAPI, PostgreSQL, and Docker**, scaling the prototype to enterprise client deployment for **L&T Technology Services**.

PROJECTS

Barely Tokenized: Decentralized P2P Smart-Grid Marketplace | *Solidity, ZoKrates, Next.js, Node.js, MongoDB* **Jan 2025 - Feb 2025** | GitHub

- **Developed** a decentralized peer-to-peer energy trading marketplace, integrating **Solidity** smart contracts, IoT meter telemetry, and **2 zero-knowledge circuits** (buyer-balance and energy-reserve proofs) for confidential trade settlement.
- **Designed** a **2-pointer** order-matching and bidding algorithm with dynamic market-cap bounds and grid-operator fallback, alongside a **Carbon Credit** scoring mechanism to quantify producer efficiency and transmission loss.
- **Built** a full-stack web application with **Next.js, Tailwind CSS, Node.js, Express, and MongoDB** for real-time consumption analytics, transaction logging, and wallet authentication, verified across **2 blockchain testnets** (Ganache and Ethereum Sepolia).

C++20 Image Engine, Huffman Codec and STL Library | *C++20, Templates, Bitwise I/O, Iterators* **Feb 2026 - Mar 2026** | GitHub-1 • GitHub-2

- **Constructed** a standalone, header-only **C++20** image library from scratch, featuring byte-aligned **24-bit BMP** parsers with row-padding and endianness handling, as well as **4 spatial convolution filters** (Sobel, Scharr, Prewitt, and Gaussian blur).
- **Implemented** a **Huffman Encoder and Decoder**, utilizing a **64 KB** buffered bit-stream reader and writer, recursive tree-header serialization, and **1-indexed** priority queues with **O(log N)** logarithmic rebalancing.
- **Created** an STL-compatible generic container package with random-access, forward, and bidirectional iterators across **6 container types** (Array, Singly, Circular, and Doubly Linked Lists, Stack, and Queue), along with a **Multicast Delegate** event dispatcher.

Router Quest: Fuel-Constrained Graph Routing Solver | *C++, Graph Algorithms, Floyd-Warshall, Heuristics* **Nov 2025 - Dec 2025** | GitHub

- **Formulated** a high-speed **C++** solver for the NP-hard, fuel-constrained pickup-and-delivery routing problem on weighted undirected graphs, enforcing precedence rules and tank-capacity bounds across intermediate refuel stations.
- **Combined** the **Floyd-Warshall O(V^3)** all-pairs shortest-path algorithm and parent-pointer tree reconstruction with a **2-step** lookahead fuel-feasibility simulator, multi-start greedy heuristics, and a standalone route verifier.

MolConvert: Multi-Format Molecular Compiler and CLI | *Python, CLI Architecture, 3D Geometry, Pytest* **Jun 2026 - Jul 2026** | GitHub

- **Programmed** a modular, command-line conversion tool that translates across **9 molecular file formats** (including PDB, SDF, Gaussian, and Z-Matrix) via a unified JSON intermediate representation.
- **Executed 3D** coordinate reconstruction from internal bond lengths, angles, and torsions using **NumPy** rotational matrices, backed by chemical valence sanitization, batch directory processing, and **376 Pytest unit tests**.

High-Throughput Streaming Candidate Ranking Pipeline | *Python, Bounded Min-Heap, TF-IDF, Gradio* **May 2026 - Jun 2026** | GitHub

- **Streamlined** a dependency-free, single-pass ranking workflow that evaluates **100,000 candidate records (487 MB)** in under **65 seconds** within **120 MB of peak RAM**, using an **O(N log K)** bounded min-heap and **TF-IDF** cosine similarity.
- **Devised** a **20-feature** scoring model and anomaly filter across **70,584** eligible applicants (flagging fabricated histories, timeline gaps, and keyword stuffing), alongside a **480-template** justification generator deployed via **Gradio** on **Hugging Face Spaces**.

SKILLS

Programming Languages: C++, C, Python, Solidity, JavaScript, TypeScript, SQL, Bash, LaTeX.
C++ and Core Systems: STL, Templates and Generic Programming, Iterators, Memory Layout and Bitwise I/O, Data Structures and Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS, Computer Networks, System Design.
Backend, Web and Cloud Tools: FastAPI, Node.js, Express.js, Next.js, React.js, Tailwind CSS, REST APIs, PostgreSQL, MongoDB, Docker, Git, Linux, Pytest, Streamlit, Gradio, Hugging Face Spaces.
Blockchain and Cryptography: Smart Contracts (EVM), Zero-Knowledge Proofs (ZK-SNARKs), Hardhat, Web3.js, MetaMask, Ethereum.
Relevant Coursework: Introduction to Computing (CS 101), Data Structures and Algorithms, Linear Algebra, Multivariable Calculus and Differential Equations, Quantum Computing (EE 696), Basic Electronics (EE 101), Hardware-Aware Deep Learning.

ACHIEVEMENTS AND LEADERSHIP

- **Selected** among the **1st delegation from IIT Guwahati** to represent the institute at the **ETHGlobal New Delhi Hackathon (Oct 2025)**, delivering Web3 protocol infrastructure during a **36-hour** international sprint.
- **Ranked** in the **Top 10 Percentile** among **1,000+ participants** nationwide in the **Summer Analytics** Machine Learning competition (**Jul 2024**, Certificate); completed **Product Matters 4.0** (Certificate).
- **Spearheaded** the Blockchain and FinTech curriculum as **Executive Member** of the Finance and Economics Club (**Mar 2025 - Apr 2026**), mentoring **200+ students** across distributed ledger and cryptography workshops.
- **Elected** to the Hostel Management Committee (**Apr 2025 - May 2026**), overseeing operations and events for **400+ residents**; wrote and directed a short film, winning **4th Place** campus-wide (Video).